FED SUP CLASS
5310

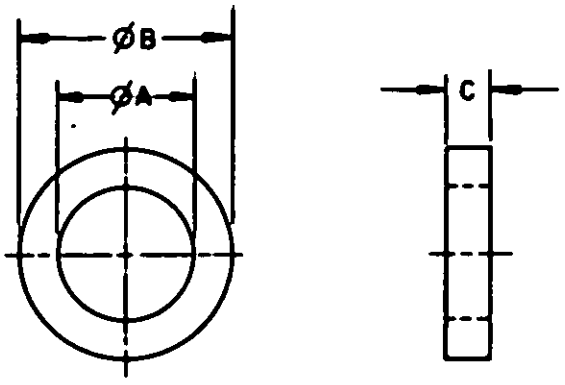



TABLE I. DASH NUMBERS AND DIMENSIONS.

| DASH NUMBERS | | | | | | NOMINAL WASHER SIZE | ØA | | ØB | | C | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ALLOY STEEL | NICKEL ALLOY | AL ALY 2024, T4 | AL ALY 7075, T6 | BRASS | CRES | | MAX | MIN | MAX | MIN | MAX | MIN |
| 115 | 88 | 01 | 21 | 41 | 61 | .060 | .068 | .063 | .130 | .120 | .028 | .022 |
| 116 | 89 | 02 | 22 | 42 | 62 | .060 | .068 | .063 | .161 | .151 | .019 | .013 |
| 117 | 90 | 03 | 23 | 43 | 63 | .073 | .099 | .089 | .161 | .156 | .028 | .022 |
| 118 | 91 | 19 | 39 | 59 | 79 | .073 | .099 | .089 | .140 | .130 | .013 | .010 |
| 119 | 92 | 04 | 24 | 44 | 64 | .099 | .109 | .104 | .219 | .214 | .028 | .022 |
| 200 | 93 | 05 | 25 | 45 | 65 | .112 | .125 | .120 | .180 | .170 | .032 | .029 |
| 201 | 94 | 06 | 26 | 46 | 66 | .112 | .125 | .120 | .255 | .250 | .019 | .013 |
| 202 | 95 | 20 | 40 | 60 | 80 | .112 | .125 | .120 | .180 | .170 | .019 | .013 |
| 203 | 96 | 121 (B) | 141 (B) | 161 (B) | 81 | .112 | .125 | .120 | .200 | .180 | .045 | .035 |
| 204 | 97 | 07 | 27 | 47 | 67 | .112 | .125 | .120 | .255 | .250 | .036 | .028 |
| 205 | 98 | 08 | 28 | 48 | 68 | .138 | .156 | .146 | .284 | .274 | .019 | .013 |
| 206 | 99 | 09 | 29 | 49 | 69 | .138 | .156 | .146 | .284 | .274 | .035 | .029 |
| 207 | 100 | 10 | 30 | 50 | 70 | .164 | .176 | .169 | .284 | .278 | .035 | .029 |
| 208 | 101 | 11 | 31 | 51 | 71 | .164 | .199 | .189 | .315 | .305 | .032 | .029 |
| 209 | 102 | 12 | 32 | 52 | 72 | .164 | .199 | .189 | .315 | .305 | .063 | .060 |
| 210 | 103 | 122 (B) | 142 (B) | 162 (B) | 82 | .190 | .220 | .210 | .400 | .385 | .050 | .040 |
| 211 | 104 | 13 | 33 | 53 | 73 | .190 | .210 | .200 | .365 | .355 | .032 | .029 |
| 212 | 105 | 14 | 34 | 54 | 74 | .190 | .210 | .200 | .365 | .355 | .063 | .060 |
| 213 | 106 | 15 | 35 | 55 | 75 | .250 | .270 | .260 | .460 | .450 | .032 | .029 |
| 214 | 107 | 123 (B) | 143 (B) | 163 (B) | 83 | .250 | .280 | .260 | .375 | .350 | .090 | .070 |
| 215 | 108 | 124 (B) | 144 (B) | 164 (B) | 84 | .251 | .280 | .260 | .375 | .350 | .028 | .022 |
| 216 | 109 | 16 | 36 | 56 | 76 | .250 | .270 | .260 | .460 | .450 | .063 | .060 |
| 217 | 110 | 17 | 37 | 57 | 77 | .250 | .288 | .278 | .840 | .830 | .032 | .029 |
| 218 | 111 | 18 | 38 | 58 | 78 | .312 | .317 | .307 | .525 | .515 | .032 | .029 |
| 219 | 112 | 125 | 145 | 165 | 85 | .375 | .421 | .401 | .630 | .620 | .055 | .045 |
| 220 | 113 | 126 (B) | 146 (B) | 166 (B) | 86 | .625 | .656 | .640 | 1.062 | 1.032 | .050 | .040 |
| 221 | 114 | 127 | 147 | 167 | 87 | .750 | .812 | .790 | 1.125 | 1.100 | .019 | .013 |

(B) DENOTES CHANGE(S)

APPROVED 30 MAY 84 REVISED (A) 20 MAY 86 (B) 7 OCT 86

USER ACTIVITIES:
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AMSC N/A

| P.A. ARMY-AR | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|
| Other Cust AIR FORCE-99 | | WASHER, FLAT (NARROW SERIES) | MS51496 |
| PROCUREMENT SPECIFICATION FF-W-92 | SUPERSEDES: | | PAGE 1 OF 2 |

DD FORM 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5310-1568

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REQUIREMENTS:

1. MATERIAL.

ALUMINUM ALLOY, 2024-T4, IN ACCORDANCE WITH QQ-A-250/5,
ALUMINUM ALLOY, 7075-T6, IN ACCORDANCE WITH QQ-A-250/12,
BRASS, COMPOSITION 11, (UNS C28000), IN ACCORDANCE WITH QQ-B-613,
CORROSION RESISTANT STEEL, TYPE 302 (UNS S30200), 304 (UNS S30400), 305 (UNS S30500), 316 (UNS S31600),
321 (UNS S32100), 347 (UNS S34700) IN ACCORDANCE WITH QQ-S-766,
NICKEL ALLOY, UNS N06600, IN ACCORDANCE WITH ASTM B168, CONDITION A, SPRING TEMPER,
ALLOY STEEL, UNS G41300, IN ACCORDANCE WITH MIL-S-6758, CONDITION (4).

2. PROTECTIVE COATING AND SURFACE FINISH.

ALUMINUM ALLOY, ANODIZE IN ACCORDANCE WITH MIL-A-8625, TYPE I OR TYPE II, CLASS 1, OR CHEMICAL SURFACE TREATMENT IN ACCORDANCE WITH MIL-C-5541, CLASS A.
BRASS - BLACK CHEMICAL FINISH IN ACCORDANCE WITH MIL-F-495.
NICKEL PLATING IN ACCORDANCE WITH QQ-N-290, CLASS 1, GRADE F.
TIN PLATING IN ACCORDANCE WITH ASTM B545, .0001 INCH THICKNESS.
CORROSION RESISTANT STEEL - CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH ASTM A380.
ALLOY STEEL - CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416 TYPE II, CLASS 2.

3. HARDNESS.

NICKEL ALLOY, CONDITION S, 30HRC MIN
ALUMINUM ALLOY, 2024-T4, 94-103HRF
ALUMINUM ALLOY, 7075-T6, 85-95HRB
BRASS, COMPOSITION 11, 1/2 HARD, 57-77HRB
CORROSION RESISTANT STEEL, 20HRC MIN.
ALLOY STEEL, 38HRC MIN

4. MAGNETIC PERMEABILITY. CORROSION RESISTANT STEEL WASHERS SHALL HAVE A MAGNETIC PERMEABILITY OF 2.0 MAXIMUM (AIR = 1.0) FOR A FIELD STRENGTH OF H = 200 OERSTEDS USING A MAGNETIC INDICATOR PER MIL-I-17214.

5. FLATNESS. WASHERS SHALL BE FLAT TO WITHIN .007 INCH.

6. PARALLELISM. WASHERS SHALL BE PARALLEL TO WITHIN .002 INCH.

7. PART NUMBER. THE PART NUMBER CONSISTS OF THE BASIC MS NUMBER, FOLLOWED BY A FINISH CODE LETTER, FOLLOWED BY A DASH NUMBER FROM TABLE I.

EXAMPLE: MS51496C01

- DASH NUMBER FROM TABLE I
- FINISH CODES
  - D - ANODIZE
  - C - CHEMICAL SURFACE TREATMENT
  - G - BLACK CHEMICAL
  - N - NICKEL
  - P - PASSIVATE
  - T - TIN
  - E - CADMIUM
- BASIC MS NUMBER

MS51496C01 - INDICATES, WASHER, FLAT-NARROW SERIES, CHEMICAL SURFACE TREATMENT, ALUMINUM ALLOY 2024-T4, .068 ID, .130 OD, .028 THICK.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. REMOVE ALL BURRS AND SHARP EDGES BEFORE FINISH.
3. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
4. REFERENCED GOVERNMENT (OR NONGOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

APPROVED 30 MAY 84 REVISED (B) FOR CHANGE(S) SEE PAGE(S) I

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| P. A. | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|
| ARMY-AR<br>Other Cust<br>AIR FORCE-99 | | WASHER, FLAT (NARROW SERIES) | MS51496 |
| PROCUREMENT SPECIFICATION<br>FF-W-92 | SUPERSEDES: | | PAGE 2 OF 2 |



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